

First record of *Bombina variegata* (L., 1758) (Anura: Bombinatoridae) from Turkey

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Abstract: *Bombina variegata* (L., 1758) is recorded in Turkey for the first time. With the record of the present study, the known easternmost distribution area of the species has been extended from Greece to Turkey. The specimens were collected from Kurtkaya-Enez (Edirne). The morphometric characters and color-pattern features of the specimens were compared with data given in the earlier literature. We observed that the specimens of the Kurtkaya population were similar to the *B. v. scabra* specimens reported in the literature.

Key words: Yellow-bellied toad, *Bombina variegata*, locality, distribution, Enez

The yellow-bellied toad, *Bombina variegata* (L., 1758), is a member of the genus *Bombina*, which has eight species. It is distributed across Austria, Belgium, Bulgaria, the Czech Republic, France, Germany, Greece, Hungary, Italy, Liechtenstein, Luxembourg, Moldova, Montenegro, the Netherlands, Poland, Romania, Serbia, Slovakia, Slovenia, Switzerland, and Ukraine (<http://amphibiaweb.org/>). It is a polytypic species, represented by four subspecies, widespread in central and southeastern Europe (Gollmann et al., 1997). The nominotypical subspecies extends from France to the Carpathian Mountains. The second subspecies, *B. v. pachypus* (Bonaparte, 1838), is confined to the Apennines, south of the Po River valley in Italy (Vukov et al., 2006). Another taxon, *B. v. scabra* (Küster, 1843), is found in the Balkans (Gollmann et al., 1997; Arnold and Ovenden, 2002). The fourth subspecies, *B. v. kolombatovici* (Bedriaga, 1890), is distributed in a restricted area of the Dalmatian coast (Szymura, 1993, 1998; Vukov et al., 2006). *Bombina variegata* has been classified in the LC (Least Concern) category of the IUCN Red List of Threatened Animals since 2004 (IUCN, 2009).

Until now, only one species (*Bombina bombina*) belonging to the genus *Bombina* has been recorded in Turkey (Başoğlu and Özeti, 1973; Baran et al., 2012). The present study introduces a new bombinatorid frog species, *Bombina variegata*, to the Turkish herpetofauna.

During the field studies, 14 adult (8 male, 6 female) *Bombina variegata* specimens were collected from the locality of Kurtkaya (vicinity of Gala Lake), Enez, Edirne, on 17 June 2015 (GPS data, 40°45'104"0, 26°13'508"0; 181

m a.s.l.). The locality is approximately 10 km from the border with Greece. The locality map is shown in Figure 1. Adult males were recognized by the presence of nuptial pads on their front legs. All specimens were anesthetized with ether, fixed with a 10% formaldehyde injection, and preserved in 70% ethanol. They were deposited in the Zoology Laboratory (Collection No. KZL-199 for Enez) of the Department of Biology of Karadeniz Technical University.

The morphometric data were recorded following the systems of Radojičić et al. (2002) and Vukov et al. (2006). All morphometric features were measured using a digital caliper with an accuracy of 0.01 mm. All measured data were compared with the studies of Radojičić et al. (2002) and Vukov et al. (2006). The following morphometric characteristics were evaluated: L, total length from the top of the snout to the edge of the cloaca; F, distance from the center of the cloaca to the knee joint; T, distance from the knee joint to the tibiotarsal ankle; N, distance from the tibiotarsal ankle to the top of the longest toe; P, distance from the metatarsal ankle to the top of the longest toe; H, distance from the shoulder joint to the elbow joint; M, distance from the elbow joint to the top of the longest finger; DpPa, length of the thumb of the front leg; DsPa, length of the second finger on the front leg; DpPp, distance from the metatarsal tubercle to the top of the thumb; Cint, largest distance of the metatarsal tubercle; Lc, head length from the top of the snout to the corner of the mouth; Ltc, head width at the angle of the jaw; Spp, shortest distance between the inner eyelids; Spi, distance between the

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Figure 1. Map showing the distribution area (from <http://amphibiaweb.org/>) belonging to four subspecies in Europe and the new record (blue star) of *Bombina variegata* for Turkey (*B. v. variegata* is represented by the orange area, whereas *B. v. pachypus*, *B. v. kolombatovici*, and *B. v. scabra* are represented by the purple, red, and green areas, respectively).

nostrils; Spcr, distance between the front end of the eyes; Lo, largest length of the eyes; Ltp, largest width of the upper eyelid; Dro, distance between the top of the snout and the front end of the eye; and Dno, distance between the nostrils and the front end of the eye.

The habitat of the specimens from Kurtkaya consists of deciduous forest (Figure 2). The vegetation of the habitat generally comprises broad-leaved trees (*Quercus ilex*, *Fraxinus* sp., *Tilia* sp., *Paliurus aculeatus*, and *Acacia* sp.). *B. variegata*, *Pelophylax bedriagae* (Camerano, 1768), and *Bufo variabilis* (Pallas, 1769) live in sympatry at the locality of Kurtkaya. The specimens were found during a day excursion between 0930 and 1400 hours. The temperature was between 28 and 33 °C.

Material: KZL-199/2015, 8 ♂♂, 6 ♀♀, 17.06.2015, Kurtkaya, Enez, Edirne leg. U. BÜLBÜL, M. KURNAZ, and A.İ. EROĞLU.

Twenty different morphological characteristics belonging to 8 male and 6 female specimens from the Kurtkaya population were evaluated. Three characteristics (L, DpPa, and Spcr) were higher in females, whereas the remaining 17 were found to be higher in males. Mean L was 42.24 (39.42–44.69) mm in male specimens and 43.00 (40.42–48.70) mm in female specimens.

The mean values and standard errors of the 20 morphological characteristics belonging to 8 male and 6 female specimens from Kurtkaya are shown in Table 1.

The coloration of *B. variegata* specimens collected from Kurtkaya was very similar in both males and females. The color of the upper surface of the body varied from olive-dark to bright brown. There were small dark spots on the dorsal surface of the body. There were greenish spots on both fore- and hind limbs. The upper surface of the body contained scattered dark tubercles. The tubercles were acute and relatively high. The dorsal warts were cone-shaped. The belly was generally yellowish, although it was orange in some specimens. In all specimens, black spots were scattered on a yellow area of the belly. The bright coloration exceeded the dark coloration on the belly and was never restricted to the dark zone. The ventral tubercles were very small and darkish. The toe tips (right and left) of the specimens were light. In contrast to the females, the males had nuptial pads on the 1st, 2nd, and 3rd fingers (right and left). Dorsal and ventral views of some specimens are given in Figure 3.

Although *Bombina variegata* is not known to be from Turkey according to the current literature, we provide here the first locality record of a *Bombina variegata* population present in Turkey. The locality (Kurtkaya, Enez) is about 10 km away from the border with Greece.

The habitat of our specimens consists of a deciduous forest on a hill. In general, *Bombina variegata* lives mainly in foothills and mountains; in coniferous, deciduous, and mixed forests; and in bushlands, meadows, floodplains,

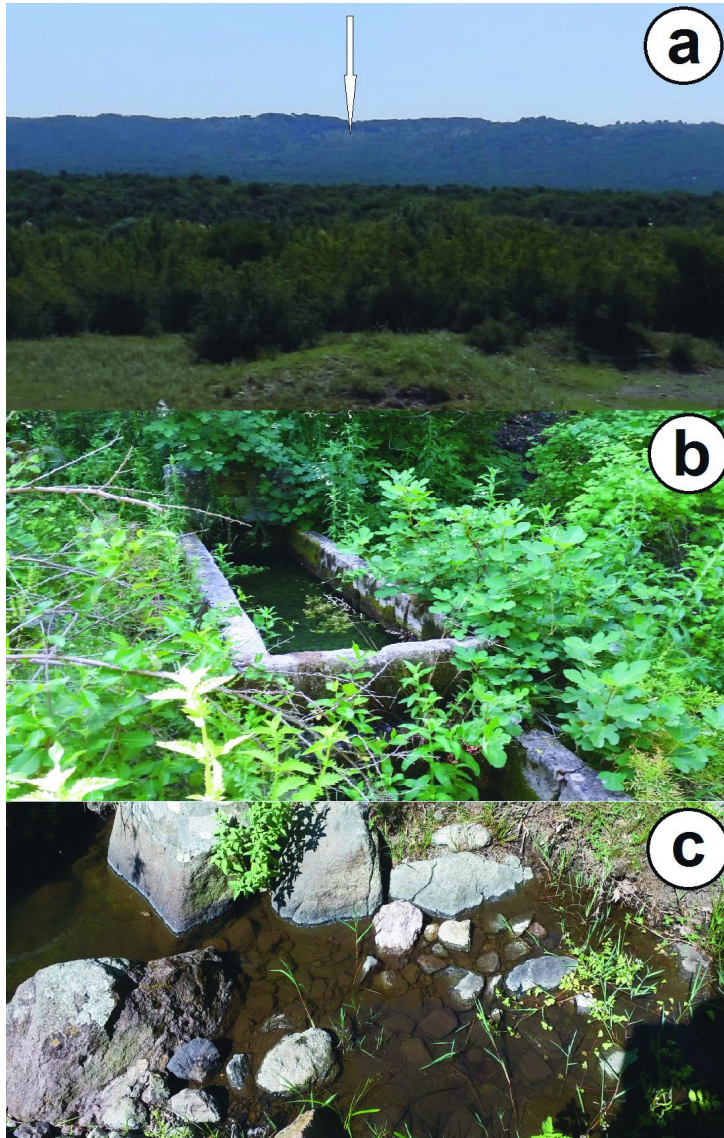


Figure 2. The habitat of *B. variegata* from Kurtkaya, Enez. (a- View of the slope, the arrow shows the place of the toads in the slope; b- View of the ponds where the toads were found; c- View of a small stream in the locality).

grasslands, etc. (Lörcher, 1969; Madej, 1973; Barandun, 1995; Kuzmin, 1999; Köhler, 2003; Denoël, 2004; Hartel et al., 2007; <http://amphibiaweb.org/>).

Based on the comparison of the color-pattern features of our specimens with the data of Stugren and Vancea (1968), we found that our specimens were similar to the *Bombina variegata* specimens used in their study.

In order to evaluate the subspecific status of our specimens, we compared their morphometric measurements to those given in previous studies (Table 2). Our morphometric data from the Kurtkaya population were considered in relation to the records of Radojičić et al. (2002) (Macedonia, Montenegro, and Serbia) and

Vukov et al. (2006) (Macedonia, Kosovo, Montenegro, and Serbia) related to the subspecies *B. v. scabra*, whose geographic distribution is very close to Turkey. We evaluated that although there were slight differences with the study of Radojičić et al. (2002), based on L, F, N, DpPp, and Lc values for males and L, F, T, N, DpPp, Lc, Ltc, and Ltp values for females, the morphometric characters of the Kurtkaya population were in agreement with the data of Radojičić et al. (2002). Similarly, although there were slight differences with the study of Vukov et al. (2006) based on L and F values for males and L and H values for females, there was similarity between the results of both studies. Although our results were found

Table 1. Descriptive statistics of some morphometric measurements of *Bombina variegata* collected from Kurtkaya. For abbreviations, see text (n: number of samples; Range: minimum and maximum value; SE: standard error).

	♂♂				♀♀				♂♂+♀♀			
Characters	n	Range	Mean	SE	n	Range	Mean	SE	n	Range	Mean	SE
L	8	39.42–44.69	42.24	0.64	6	40.17–47.70	43.00	1.15	14	40.42–48.70	42,57	0.60
F	8	15.11–18.58	16.48	0.45	6	14.51–18.55	16.25	0.61	14	14.51–18.58	16,38	0.39
T	8	14.39–16.74	15.93	0.29	6	13.95–16.55	14.95	0.37	14	13.95–16.74	15.51	0.26
N	8	24.53–28.97	26.93	0.50	6	23.73–29.90	25.96	0.90	14	23.73–29.90	26.51	0.53
P	8	14.94–17.82	16.56	0.36	6	13.34–18.02	16.29	0.70	14	13.34–18.02	16.44	0.47
H	8	9.18–11.66	10.37	0.24	6	7.84–8.73	8.28	0.16	14	7.84–11.66	9.47	0.21
M	8	15.20–17.60	16.64	0.31	6	14.90–17.79	15.86	0.47	14	14.90–17.79	16.31	0.28
DpPa	8	3.16–4.96	4.12	0.19	6	4.20–5.91	4.84	0.30	14	3.16–5.91	4.42	0.17
DsPa	8	5.88–7.38	6.75	0,18	6	6.10–7.40	6.59	0.21	14	5.88–7.40	6.68	0.13
DpPp	8	2.45–4.69	3.80	0.27	6	2.92–4.25	3.68	0.21	14	2.45–4.69	3.75	0.17
Cint	8	1.47–2.30	1.75	0.09	6	1.16–1.77	1.41	0.10	14	1.16–2.30	1.61	0.08
Lc	8	8.61–10.77	9.88	0.28	6	8.41–10.16	9.38	0.26	14	8.41–10.77	9.67	0.20
Ltc	8	11.87–14.36	13.39	0.32	6	11.98–14.33	12.70	0.39	14	11.87–14.36	13.09	0.26
Spp	8	2.34–3.32	2.72	0.12	6	2.12–2.86	2.51	0.12	14	2.12–3.32	2.63	0.08
Spi	8	2.37–3.36	2.75	0.12	6	2.30–3.06	2.59	0.11	14	2.30–3.36	2.68	0.09
Spqr	8	4.98–6.16	5.62	0.14	6	5.40–6.39	5.85	0.18	14	4.98–6.39	5.72	0.15
Lo	8	3.01–3.91	3.61	0.11	6	3.01–3.85	3.54	0.13	14	3.01–3.91	3.58	0.09
Ltp	8	2.15–2.91	2.57	0.09	6	1.96–2.53	2.21	0.10	14	1.96–2.91	2.41	0.08
Dro	8	4.44–5.26	4.77	0.12	6	4.08–5.25	4.62	0.17	14	4.08–5.26	4.71	0.10
Dno	8	2.79–3.78	3.15	0.11	6	2.41–3.01	2.75	0.11	14	2.41–3.78	2.98	0.09

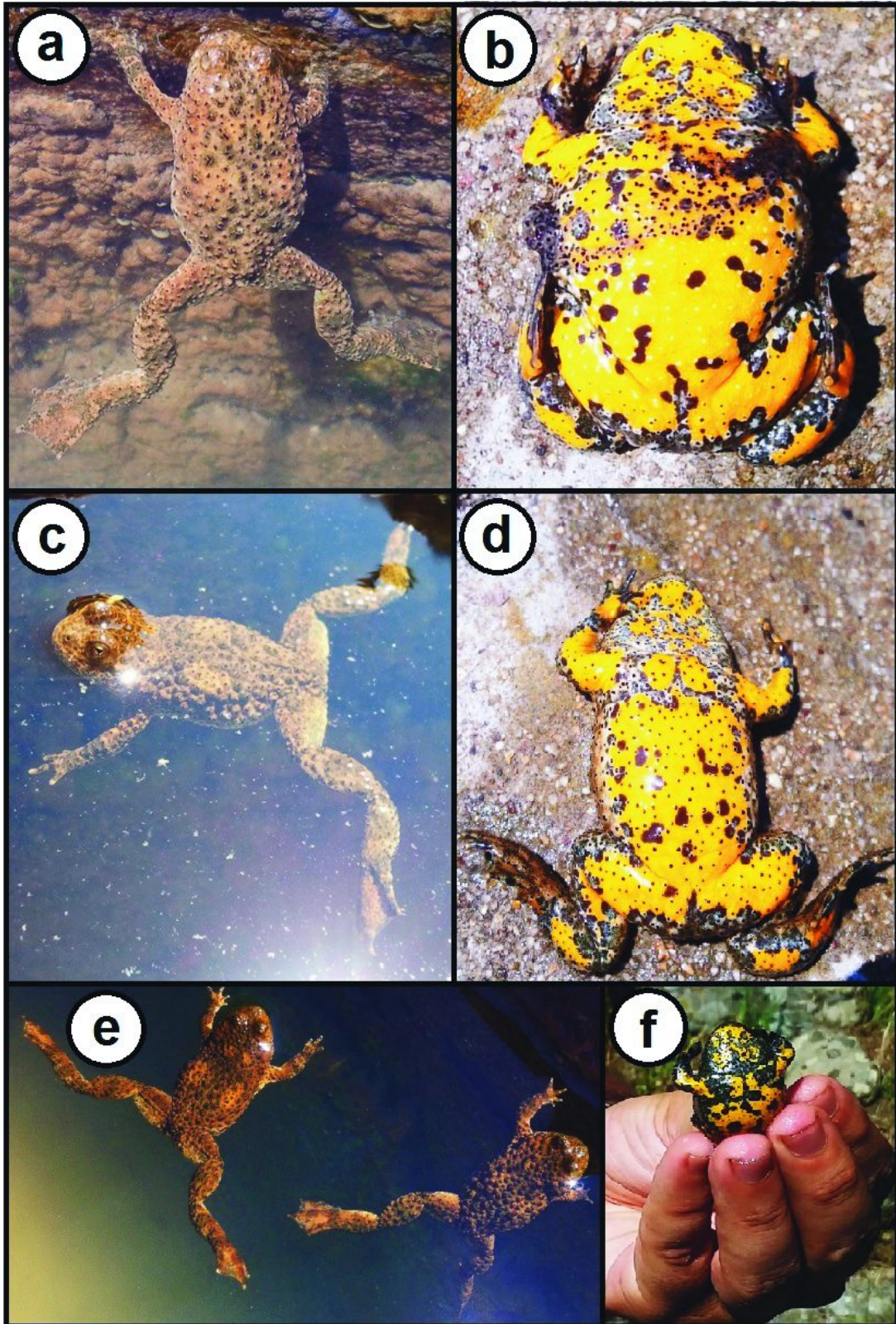


Figure 3. General views of the toads in their natural habitat. (a- Dorsal view of a female specimen; b- Ventral view of a female specimen; c- Dorsal view of a male specimen; d- Ventral view of a male specimen; e- The toads swimming in their breeding pond; f- Unken reflex of a male specimen).

Table 2. Comparison of morphometric measurements of our specimens with those given by Radojičić et al. (2002) and Vukov et al. (2006) for *B. v. scabra*. For abbreviations, see text (n: number of specimens).

	Radojičić et al. (2002)				Vukov et al. (2006)				This study			
	♂♂		♀♀		♂♂		♀♀		♂♂		♀♀	
Characters	n	Mean	n	Mean	n	Mean	n	Mean	n	Mean	n	Mean
L	111	43.50	87	44.60	213	43.64	140	44.61	8	42.24	6	43.00
F	111	17.70	87	17.70	213	17.09	140	16.54	8	16.48	6	16.25
T	111	16.80	87	16.20	213	16.19	140	15.37	8	15.93	6	14.95
N	111	25.20	87	24.80	213	26.82	140	25.84	8	26.93	6	25.96
P	111	16.60	87	16.10	213	17.16	140	16.50	8	16.56	6	16.29
H	111	10.50	87	9.20	213	10.48	140	9.04	8	10.37	6	8.28
M	111	16.50	87	16.50	213	16.96	140	16.56	8	16.64	6	15.86
DpPa	111	4.20	87	4.20	213	4.60	140	4.52	8	4.12	6	4.84
DsPa	111	6.30	87	6.30	213	6.62	140	6.55	8	6.75	6	6.59
DpPp	111	5.40	87	5.40	213	4.29	140	4.20	8	3.80	6	3.68
Cint	111	1.40	87	1.40	213	1.36	140	1.35	8	1.75	6	1.41
Lc	111	8.10	87	8.00	213	10.16	140	10.02	8	9.88	6	9.38
Ltc	111	14.20	87	14.30	213	13.53	140	13.30	8	13.39	6	12.70
Spp	111	2.60	87	2.60	213	2.78	140	2.70	8	2.72	6	2.51
Spi	111	2.40	87	2.40	213	2.59	140	2.51	8	2.75	6	2.59
Spcr	111	5.40	87	5.30	213	6.08	140	6.01	8	5.62	6	5.85
Lo	111	3.40	87	3.40	213	3.58	140	3.60	8	3.61	6	3.54
Ltp	111	3.10	87	3.10	213	2.72	140	2.72	8	2.57	6	2.21
Dro	111	5.00	87	4.80	213	5.01	140	4.91	8	4.77	6	4.62
Dno	111	3.20	87	3.10	213	3.14	140	3.06	8	3.15	6	2.75

similar to the records of Radojičić et al. (2002) and Vukov et al. (2006), the number of the specimens in our study was very low. More specimens should be investigated to evaluate the similarity of the Kurtkaya population with the Balkan populations of *B. v. scabra*. Based on our findings, geographic distributions of the subspecies of *B. variegata*, and morphometric data from the studies of Radojičić et al. (2002) and Vukov et al. (2006) belonging to *B. v. scabra*, we conclude that our specimens belong to the *B. v. scabra*

subspecies. However, our conclusion is not dependent on molecular data.

B. v. scabra was separated from the nominate subspecies by bioacoustic, electrophoretic, and mitochondrial genome size data (Vasara et al., 1991; Szymura et al., 2000; Borkin et al., 2005). Previous molecular data, based on mtDNA, produced 2 Balkan *B. v. scabra* lineages representing the Balkano-Western haplogroup (Pabijan et al., 2013).

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